

TRANSFERABILITY OF DEEP LEARNING-BASED WILDLIFE CLASSIFICATION ACROSS SITES

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Engineering and
Physical Sciences
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Problem

- This project focuses on **single-species classification of camera-trap image data** collected across multiple sites in the Kruger National Park, South Africa.
- A subset of these images from both 2015 and 2016 have been manually annotated, with the **labels for the 2016 being unverified**.
- All 2015 sites are artificial waterholes, whereas **2016 sites are a combination of artificial and natural waterholes**.
- We investigate the **performance of the model when transferring between these sites**.
- Due to data scarcity, **duplication of images was necessary** to meet training and test set quotas, especially for species which are underrepresented in this pool.

Dataset and Methodology

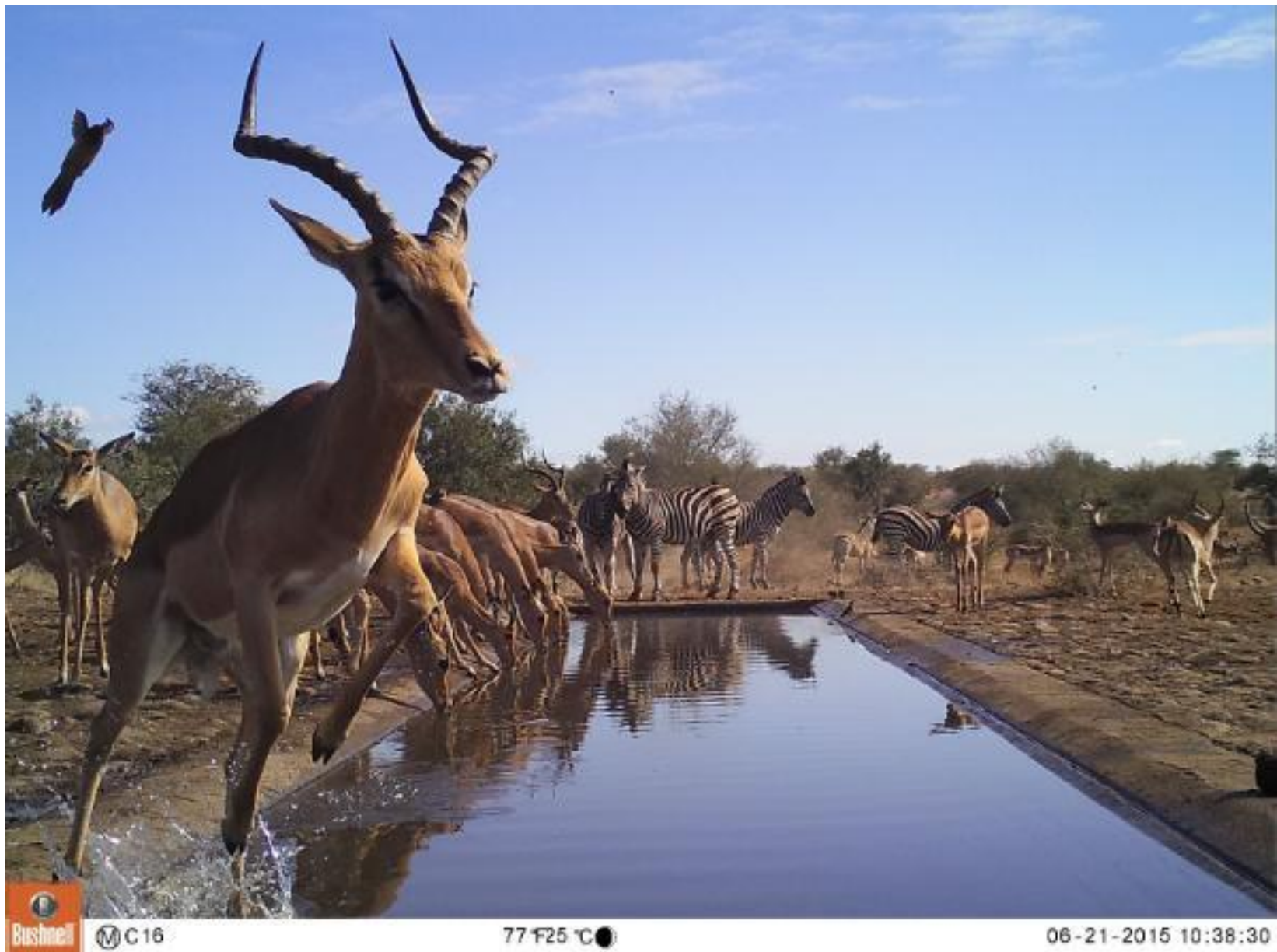


Figure 1: Example 2015 camera-trap image [1]



Figure 2: Example 2016 camera-trap image

After cleaning the data, we set a **threshold of 300 total images per class**. This left **8 species classes**.

We used balanced per-species quotas with different splits across 2015/2016 for training images (**600 total per species**): **2015 images only**, a **1:1 ratio** of 2015 to 2016 images, and a **5:1 ratio** of 2015 to 2016 images.

Training set-up:

- ResNet18**, pretrained on ImageNet, **fully fine-tuned**, with data augmentation, [2]
- Adam optimiser** with **cross entropy loss function**,
- Fixed random seed for reproducibility.

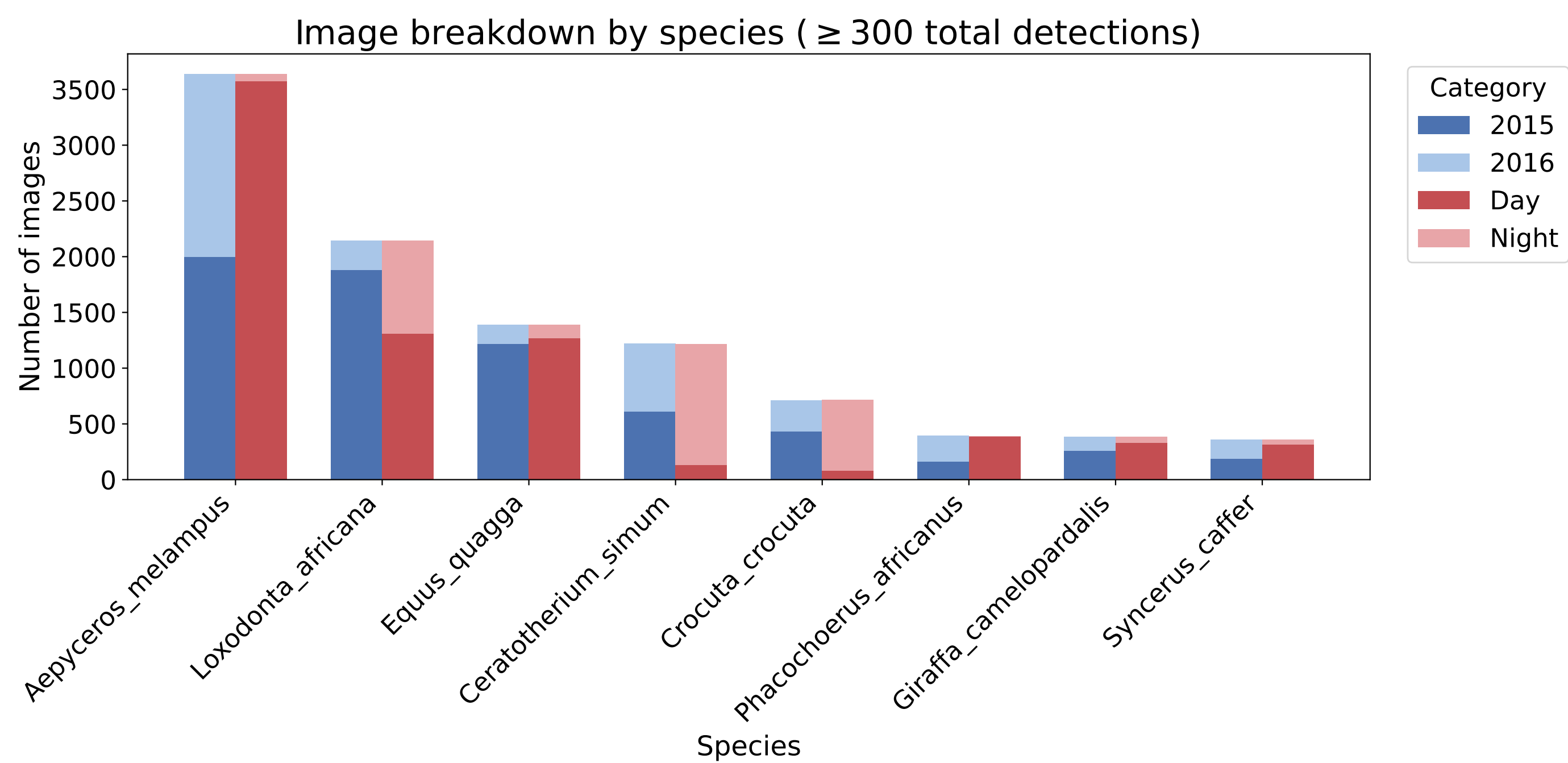


Figure 3: Species distributions by category

Evaluation

Binomial	Species	2015	2016	Day	Night
Aepyceros melampus	Impala	500	410	892	18
Loxodonta africana	Elephant	470	65	309	226
Equus quagga	Zebra	304	43	317	30
Ceratotherium simum	Rhino	152	152	31	273
Crocuta crocuta	Hyena	108	70	19	159
Phacochoerus africanus	Warthog	40	58	97	1
Giraffa camelopardalis	Giraffe	64	32	88	8
Syncerus caffer	Buffalo	46	43	67	22

Table 1: Test support per species breakdown

Results: Best overall test accuracy: 75.2%; best macro F1: 0.663 (achieved in the same 1:1 split experiment).

Split	2015 accuracy	2016 accuracy	Macro avg F1
2015-only (600-0)	0.761 $\pm$ 0.010	0.520 $\pm$ 0.057	0.587 $\pm$ 0.018
1:1 (300-300)	0.759 $\pm$ 0.010	0.699 $\pm$ 0.021	0.653 $\pm$ 0.011
5:1 (500-100)	0.764 $\pm$ 0.006	0.685 $\pm$ 0.027	0.650 $\pm$ 0.009

Table 2: Overall results (mean $\pm$ std over 3 seeds) by training split

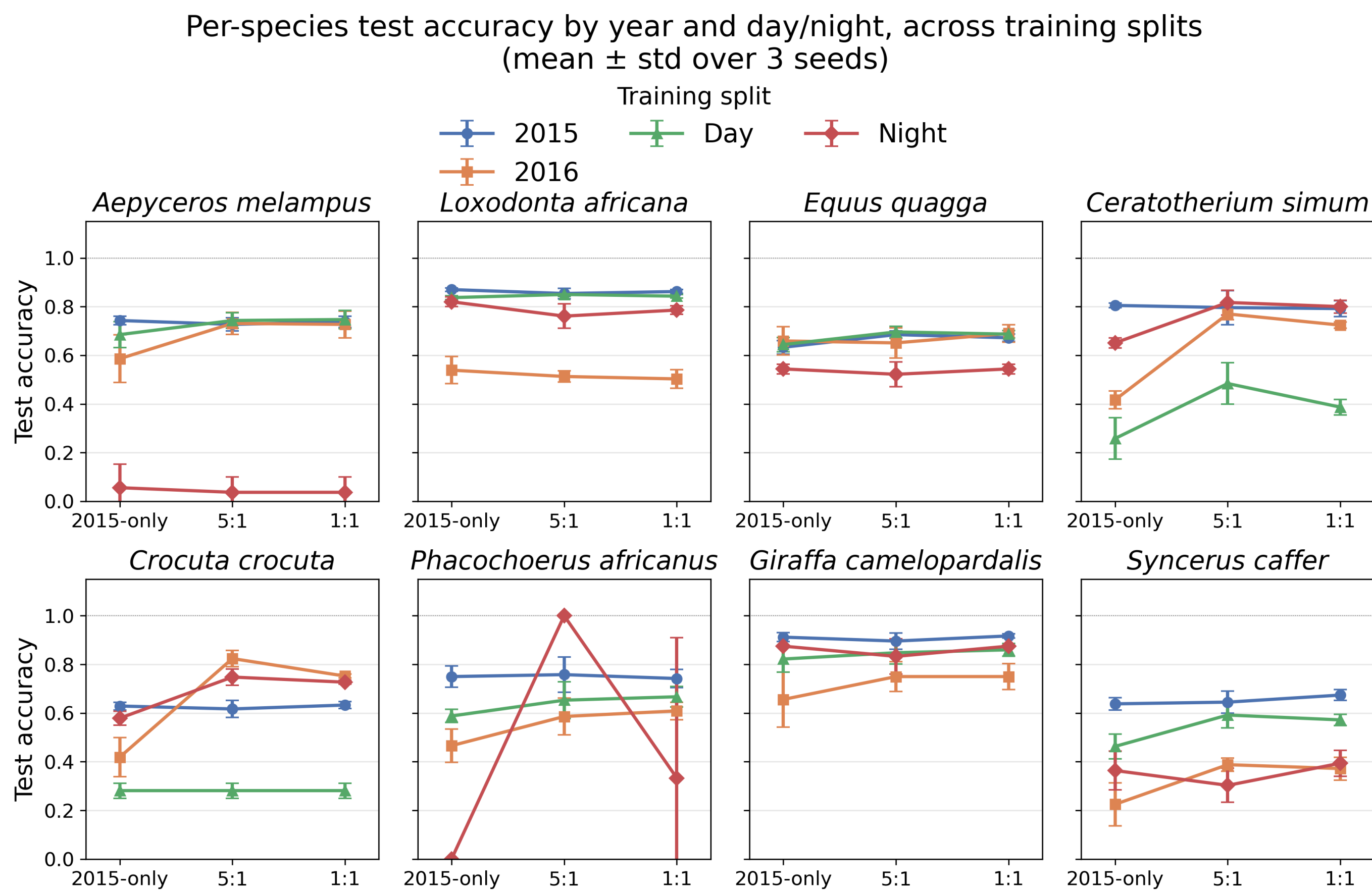


Figure 4: Results breakdown by species

Conclusions

- Since 2015 and 2016 sites don't overlap, we can see that the model often doesn't transfer well between sites, and the **best results** were achieved when **2016 images were included in the training set**.
- The steady results for 2015 throughout and the comparable results between the 1:1 and 5:1 splits suggest that a **smaller inclusion of both 2015 and 2016 images is still sufficient to achieve high performance**.
- From our results, we see that the **5:1 split would be the best to use in practise** as the model performs only slightly worse than the 1:1 split but uses less of the unreliable 2016 images.
- We see that the **model performs better in the day/night** depending on when that **species is more active**, as there is more data for that species at that time.
- We also note the **volatile and low performing results due to data scarcity** in some species' categories.

References

- Maxwell J. Farrell et al. "Environmental DNA as a management tool for tracking artificial waterhole use in savanna ecosystems". In: [Insert Conference Name, e.g., Proceedings of the International Ecology Symposium]. [Insert Organization/Publisher if applicable]. 2022, p. 109712. DOI: 10.1016/j.biocon.2022.109712. URL: <https://doi.org/10.1016/j.biocon.2022.109712>.
- Kaiming He et al. "Deep residual learning for image recognition". In: Proceedings of the IEEE conference on computer vision and pattern recognition. 2016, pp. 770–778.